Product Manual | VSG-Mini-6

USB Vector Signal Generator

Table of Contents

OVERVIEW

The VSG-Mini-6 is a compact USB vector signal generator that delivers laboratory-grade RF performance from 9 kHz to 6 GHz. It pairs a high-resolution 16-bit DAC with an FPGA-based interpolator for precise baseband synthesis, and supports a 100 MHz playback bandwidth alongside 50 MHz real-time streaming.

The FPGA interpolator provides fine, granular control over sample rate while minimizing host CPU load, delivering consistent, high-efficiency performance in demanding test workflows. Its compact form factor and single-cable USB-C connection simplify integration into automated test systems without compromising RF integrity.

A streamlined, well-documented API supports rapid development across C/C++, C#, Python, and MATLAB, with integration paths for Qt, LabVIEW, and GNU Radio. The instrument and its software run on both Windows and Linux, and on ARM and x86 processors, giving engineering teams flexibility across cross-platform system architectures.

Key Features

- Frequency range: 9 kHz to 6 GHz
- Modulation bandwidth: 100 MHz playback, 50 MHz real-time
- Built-in memory depth: 125 MB (32 M samples)
- SSB phase noise: −124 dBc/Hz at 10 kHz offset (1 GHz carrier)
- Maximum output power: 7 dBm to 14 dBm; 25 dBm (Option 10)
- Minimum output power: ≤−100 dBm across the full frequency range
- Harmonics: ≤−50 dBc (0 dBm output, typical)
- 16-bit DAC for high-quality baseband waveforms
- FPGA-based high-performance interpolator
- Fine sampling rate and reference-frequency adjustment
- Built-in GNSS for timing, positioning, and frequency calibration
- USB 3.0 / 2.0 Type-C interface
- ARM and x86 processor support; Windows and Linux support

SGSTUDIO CONTROL SOFTWARE

SGStudio is the control application for the VSG-Mini-6. It consolidates RF parameter control, mode selection, mode parameters, and real-time status into a single main window, organized into five functional areas: menu bar, RF parameter settings, mode selection, mode parameter settings, and status information. The unified layout shortens the learning curve and gives operators a workflow comparable to traditional benchtop instruments.

RF Frequency and Power Sweep Control

SGStudio supports single-point, frequency-sweep, and power-sweep modes. Operators can run automated frequency-response and compression-point characterization directly from the interface, without writing custom control code.

Real-Time IQ Playback

The instrument supports continuous real-time waveform streaming from the host or buffered playback from on-device memory. Playback mode provides higher signal bandwidth and runs without host intervention, fully offloading host processing. In replay mode, the maximum waveform length is bounded by the internal memory capacity.

SIGNAL GENERATION MODES

Digital Modulation (Option 73)

An extensive modulation library from fundamental ASK, FSK, and PSK through high-order 1024QAM. Native support for DSSS and OFDM targets modern wireless standards for field testing and R&D verification.

AM / FM

Classic analog modulation with high-precision control of modulation frequency, depth, and frequency deviation for traditional receiver testing and analog system verification.

Pulse

Integrated pulse generation with configurable pulse repetition period, duty cycle, and pulse width for timing-sensitive, pulsed-RF, and radar-signal simulation.

Multitone

Composite signals with configurable tone count, tone spacing, and phase mode, plus a definable in-band notch. Suited to dynamic-range, intermodulation-distortion, and flatness testing.

Digital Ramp Sweep

Linear frequency-modulated (chirp) generation with flexible sweep width, sweep time, and sweep period for radar simulation, wideband excitation, and frequency-response testing.

Medium Power Output (Option 10)

An optional medium-power RF port delivering up to 25 dBm from 10 MHz to 6 GHz for applications requiring higher excitation levels, such as device testing and joint debugging.

SPECIFICATIONS

Frequency

Parameter	Condition	Specification
Frequency range	—	9 kHz to 6 GHz
Frequency resolution	Output frequency < 500 MHz	1 Hz
	500 MHz ≤ output frequency ≤ 6 GHz	0.1 Hz
LO switching time	—	≤100 µs pre-programmed; ≤50 ms software controlled
Reference clock	—	Internal or external; manual correction or GNSS calibration available
Frequency accuracy	TCXO (standard)	<0.5 ppm, manual correction available
	OCXO (Option 01)	<0.2 ppm, manual correction available
	OCXO correction via GNSS	≤0.05 ppm, when GNSS is locked
Aging & temperature stability	TCXO (standard)	≤1 ppm/year, ≤1 ppm
	OCXO (Option 01)	≤1 ppm/year, ≤0.15 ppm
Built-in GNSS 1PPS accuracy	—	±100 ns

Spectrum Purity

SSB Phase Noise (dBc/Hz)

Offset from the carrier	1 GHz	3 GHz	6 GHz
1 kHz	−115	−105	−98
10 kHz	−124	−114	−108
100 kHz	−125	−116	−110
1 MHz	−138	−128	−122

Harmonics (CW, 0 dBm)

Carrier frequency	Specification
100 MHz	≤−45 dBc
1 GHz	≤−50 dBc
3 GHz	≤−60 dBc
6 GHz	≤−75 dBc

Non-harmonic Spurious & EVM

Parameter	Condition	Specification
Non-harmonic spurious	1 MHz step, 20 MHz observation BW	≤−80 dBc for ~98% of frequency points; worst-case ≤−55 dBc
EVM (typical)	1 GHz	≤0.3% (1 MSPS QAM 16, α = 0.35); ≤0.5% (10 MSPS QAM 64, α = 0.35)
	6 GHz	≤0.5% (1 MSPS QAM 16, α = 0.35); ≤1.0% (10 MSPS QAM 64, α = 0.35)

Amplitude

Output Power vs. Frequency

Parameter	9 kHz	100 MHz	1 GHz	3 GHz	6 GHz
Max. output power (dBm)	≥0	≥7	≥14	≥14	≥7
Min. output power (dBm)	≤−100	≤−100	≤−100	≤−100	≤−100

Power Accuracy & Step Size

Condition	Guaranteed / Typical
Output power ≥ −45 dBm	±1.2 dB / 0.7 dB
Output power −80 dBm to −45 dBm	±1.5 dB / 1.2 dB
Output power −100 dBm to −80 dBm	±2.0 dB / 1.8 dB
Power setting step size	0.1 dB

Signal Processing

Parameter	Specification
Standard functions	Single-tone, multitone, frequency sweep, power sweep, AM, FM, digital ramp sweep, AWGN, IQ playback, real-time IQ playback
Modulation bandwidth	100 MHz RAM playback; 50 MHz continuous streaming
Built-in memory depth	125 MB (32 M samples)
IQ sampling rate	195.3125 kHz to 125 MHz, step size ≤10 Hz
Basic modulation generation	APSK: 16APSK ASK: 2ASK, 4ASK, 8ASK FSK: 2FSK, 4FSK, 8FSK, 16FSK QAM: 16 QAM, 64 QAM, 256 QAM, 1024 QAM PSK: BPSK, QPSK, 8PSK, 16PSK, DBPSK, DQPSK, D8PSK, π/4 DQPSK

General

Parameter	Specification
Power	Type-C, PD protocol (12 V / 3 A standard); voltage range 9 to 12 V; ripple <200 mVpp

Parameter	Specification
Data	Type-C, USB 3.0 (USB 2.0 bandwidth limited); requires 5 V / 1 A power supply
RF output	N(F), output impedance 50 Ω
External reference clock input	MMCX(F), amplitude ≥1.5 Vpp, input impedance 200 Ω
Reference clock output	MMCX(F), output impedance 50 Ω, 100 MHz
External trigger input	3.3 V CMOS, input high impedance
External trigger output	3.3 V CMOS
GNSS antenna input	SMA (F)
Power consumption	≤16 W
Overall/core weight	≤360 g / ≤120 g
Overall / core dimensions (L × W × H)	≤163 × 66 × 37 mm / ≤63 × 60 × 15 mm
System requirements	Linux (aarch64, x64); Windows (x64)
Operating/storage temperature	T0 class (standard): 0 to 50 °C / −20 to 70 °C T1 class (Option 40): −20 to 65 °C / −40 to 85 °C T2 class (Option 41, core only): −40 to 65 °C / −40 to 85 °C
Packaging accessories	Flash disk ×1, USB 3.0 data cable ×1, USB power cable ×1, power adapter ×1

Medium Power Output (Option 10)

Parameter	Condition	Specification
Frequency range	—	10 MHz to 6 GHz
Max. output power	—	≥25 dBm, typical
Min. output power	—	≤−80 dBm, typical
Power accuracy (Guar. / Typ.)	Output power ≥ −20 dBm	±1.2 dB / 0.7 dB
	Output power −60 dBm to −20 dBm	±1.5 dB / 0.7 dB
	Output power −80 dBm to −60 dBm	±2.5 dB / 1.2 dB
Harmonics (CW, 0 dBm)	100 MHz	≤−55 dBc
	1 GHz	≤−50 dBc
	3 GHz	≤−42 dBc
	6 GHz	≤−46 dBc

Specifications apply after a 10-minute warm-up from power-on, at an ambient temperature of 25 °C (instrument temperature 50 °C), and with adequate cooling to keep both ambient and core temperatures within the rated range. Specifications not listed for the medium-power output port are the same as the standard RF output port.

Options

Code	Description	Type
01	Built-in OCXO reference clock	Built-in hardware

Code	Description	Type
05	Built-in high-precision GNSS	Built-in hardware
10	Medium-power RF output	Built-in hardware
40	T1 temperature class	Built-in hardware
41	T2 temperature class (core only)	Built-in hardware
73	Digital modulation signal output	Software/license